

Isolation PBMCs leucosep_protocol

Work sterile in the flow cabinet

Procedure:

1. Collect the blood samples in tubes with sodium heparinate
2. Order the blood donors from low (1) to high (31) donor number and make 2 batches
3. Study day: Fill the Leucosep® tube with the separation medium: **15 mL** Ficoll plaque plus RT--> 2 tubes per donor → Ficoll has to be at least Room temperature before using leucosep tubes for PBMC isolation.
4. Close the tubes containing the Ficoll with the screw-cap and centrifuge for **30 seconds at 1000 x g at RT**. The separation medium is now located below the porous barrier.
5. Add **9 ml** of PBS to the leucosep tubes and make sure the PBS is at RT before adding blood.
6. The tubes are now ready for filling with anticoagulated blood.
We use blood:PBS in 2:1 dilution --> Poor the 2 blood tubes per donor into the correct leucosep tubes. One tube contains 8-10 ml so in total we have +/- 30 ml (2 tubes of blood + 9ml PBS)
7. Centrifuge 10 minutes at 1000 x g at RT in a swinging bucket rotor. **Switch-off brakes of the centrifuge (deceleration =1, acceleration =3).**
8. After centrifugation, the sequence of layers occurs as follows (seen from top to bottom):
 - a) Diluted Plasma
 - b) Enriched fraction (interphase of PBMC's)
 - c) Separation medium
 - d) Porous barrier
 - e) Separation medium
 - f) Pellet (erythrocytes and granulocytes)
9. Harvest the enriched cell fraction (PBMC's) by means of a sterile 1ml pipette. Collect the PBMCs into a correctly labelled 50 ml tube
10. Wash the PBMCs by adding RPMI1640 up to 35 ml and centrifuge for 7' 250 x g
11. Poor the supernatant into a waste bottle and resuspend the PBMCs by flicking the tube to loosen the pellet
12. Repeat washing step twice; resuspend the cell pellet up to a total volume of 10mL with RPMI1640 .
13. After three times washing resuspend the PBMCs in **2 ml of RPMI-1640** medium and **count (see NOBLE protocol)**, before they are ready to use for staining, cell culture or for cryopreservation

Prepare freezing media and cryovials

- Cryovials (write them with permanent writer: name, date, concentration (5-10·10⁶cells/ml) and content)
- Prepare the media in the tubes

Medium A: 100% FCS

Medium B: 20% DMSO & 80% FCS

(Final concentrations: 10% DMSO & 90% FCS)

- Store the tubes at 4°C
1. For cryopreservation of PBMCs all the supernatant has to be removed after the third wash of the PBMCs. Or wash once more 7' 250 x g
 2. Cells are resuspended in 1 ml per 10x10⁶ PBMCs ice-cold medium A (100% FCS) (10 x 10⁶ PBMCs/ml in their 15 ml tube → do this for all donors of one batch
 3. Per donor add equal amount (ml) as in point 2. of ice-cold medium B (20% DMSO & 80% FCS) is added to tube while gently mixing by shaking the tube
 4. The 1ml of cells suspension are divided over maximum of 4 (correctly labelled) cryovials (if no 4 cryovials can be filled → write down how many vials we have)
 5. Do this for all donors of the batch
 6. Put the cryovials in a white box → surround with paper towel and put is in a big Styrofoam box → put the entire box with lid in a -80 °C freezer
 7. Place the freezing container in -80°C freezer for 8-12h (overnight)
 8. Transfer the cells to liquid nitrogen or the -160 °C